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Prescribed-Curvature-Distribution Airfoils for the Preliminary Geometric Design of Axial-Turbomachinery Cascades

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Blade surfaces with continuous curvature and continuous slope of curvature minimize the possibility of flow separation, lead to improved blade designs, and reduce the direct and inverse blade-design iterations for the selection of isolated airfoils and gas-turbine-blade cascades. A method for generating two-dimensional blade shapes is presented. The geometry near the trailing edge is specified by an analytic polynomial, the main portion of the blade surface is mapped using as input a prescribed surface-curvature distribution, and the leading edge is specified as a thickness distribution added to a construction line. This procedure is similar for the suction and pressure surfaces, and by specification it constructs continuous slope-of-curvature surfaces that result in smooth surface-Mach-number and surface-pressure distributions. The method can be used to generate subsonic or supersonic airfoils for compressors and turbines, or isolated airfoils. The resulting geometric shapes can be used as inputs to various blade-design sequences. It is shown that, with other cascade-design parameters being equal, increasing the stagger angle of turbine blades results in more front-loaded and thinner blades, and that there is an optimum stagger angle resulting in minimum wake thickness. The subsonic axial-turbine blade rows included for discussion in this paper have been designed by iterative modifications of the blade geometry to obtain a desirable velocity distribution. The blade-design method can be used to improve the aerodynamic and heat transfer performance of turbine cascades, and it can result in high-performance airfoils, even if using the direct method exclusively, in very few iterations.

Introduction

Gas turbine blades are three-dimensional objects operating in a complex and unsteady flow field. Due to its complexity, the blade-design problem is usually reduced to a series of "stacked" two-dimensional problems, assuming that the cascade operates in a steady two-dimensional flow field. Blade design is partly a science and partly an art. The performance of the blades is expressed in the form of some property distribution (such as pressure, velocity, etc.) on the blade surface and in the cascade passage. Three-dimensional turbine blades are generated by "stacking" two-dimensional designs using rules for the flow angles from hub to tip at the leading and trailing edge, derived from radial-equilibrium throughflow calculations; the locus of the centers of gravity of the sections; and the three-dimensional shape of the leading edge and the trailing edge. Compromises in performance must be made to accommodate these three-dimensional constraints, the location of cooling passages, and hollow sections.

Various investigators use different definitions for the terms direct method and inverse, semi-inverse, full-inverse, or full-optimization methods (Meauze, 1989), analysis and design modes (Stow, 1989), optimization and design methods (Bry, 1989), and others, for the approaches and methods used in turbine design sequences. For the purposes of this paper we *define* as direct the method in which the designer inputs the geometry of the cascade and the output is the performance (from an analysis code) in terms of surface Mach number and pressure distributions. The performance provides guidelines for where to increase or decrease the loading, and how to modify the surface geometry in successive iterations. This means iterative use of a geometric description package and of an analysis code until a desirable performance is obtained from the analysis code. Also for the purposes of this paper we *define* as inverse the various methods in which the designer specifies the performance of the cascade to obtain the geometry, or modifications to a portion of the surface velocity or pressure distribution to obtain modifications to the geometry. This latter definition includes what other investigators define as fully inverse, semi-inverse, or simply design methods.

Both direct and inverse methods are based on the assumption of steady flow conditions, and require numerous iterations until a desirable blade shape and performance are obtained.

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The direct method is laborious, and it requires considerable insight. On the other hand, the designer has direct control of the various geometric parameters, and structurally or dynamically infeasible blade shapes are excluded before they are analyzed. One would like to start the inverse methods with a blade shape close to an acceptable design, since this will enormously reduce the number of inverse-design iterations. Therefore the designer needs to start inverse-design iterations from a good blade shape obtained by a direct method. The inverse method is less laborious, but many (although not all) inverse-design programs must start from a blade geometry generated by a direct method. The designer has less control on the blade shape (by the very purpose of inverse design). It can result in blades that are structurally undesirable due to stress considerations or location of cooling passages, or are impossible to manufacture. If one decides that some geometric parameters (such as stagger angle) must be changed, one must start from the beginning (direct method) again. The final design is usually obtained by a judicious combination of both methods.

Surface-Curvature Effects

Both the curvature and the slope of curvature of blade surfaces affect boundary-layer development and blade performance. The slope-of-curvature effects are less obvious than the curvature effects. Theoretical and experimental evidence of their importance is presented below.

Theoretical Evidence. The flow in turbomachinery cascades *curves* around the blades. The dependence of flow on curvature is seen if one writes the compressible-flow Navier–Stokes equations in the limiting case of cylindrical coordinates (r, θ, z). In particular, since we are interested in "stacking" two-dimensional solutions and geometries, we present here the equations of conservation of mass and momenta in polar coordinates (r, θ), taken from the text by Bird et al. (1960):

$$\frac{\partial \rho}{\partial t}+\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r)+\frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta)=0 \tag{1}$$

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}\right)= -\frac{\partial p}{\partial r}-\left(\frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rr})+\frac{1}{r}\frac{\partial \tau_{r\theta}}{\partial \theta}-\frac{\tau_{\theta\theta}}{r}\right)+\rho g_r \tag{2}$$

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r v_\theta}{r}\right)= -\frac{1}{r}\frac{\partial p}{\partial \theta}-\left(\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\tau_{r\theta})+\frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta}\right)+\rho g_\theta \tag{3}$$

where in the above equations p is the static pressure, and for Newtonian fluids the stress components τ_{ij} are related to the dynamic viscosity μ by the equations:

$$\begin{aligned}
\tau_{rr}&=-\mu\left[2\frac{\partial v_r}{\partial r}-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]\\
\tau_{\theta\theta}&=-\mu\left[2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right)-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]\\
\tau_{r\theta}&=\tau_{\theta r}=-\mu\left[r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta}\right]
\end{aligned} \tag{4}$$

$$(\nabla\cdot\mathbf{v})=\frac{1}{r}\frac{\partial}{\partial r}(rv_r)+\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}$$

A similar dependence of incompressible flow on curvature is also shown in most fluid dynamics texts in cylindrical coordinates, and Schlichting (1960, p. 112) attributes a set of similar equations for incompressible flow in curvilinear coordinates (x, r, y) to W. Tollmien. The $1/r$ and $1/r^2$ terms in these equations suggest a strong dependence of local pressure and velocity on curvature. Turbine-blade performance depends on the behavior of the boundary layer. While one expects that the boundary layer shields the core of the flow from surface discontinuities, the boundary-layer behavior depends on the slope of the velocity distribution as it approaches the solid wall, and Eqs. (4) indicate that the boundary layer and its behavior are affected by the local radius. Smooth velocity

Nomenclature

A = flow-area distribution along x or X (Fig. 4*b*)
b = axial chord length (nondimensionally $b=1$)
c = thickness-distribution coefficients (Eqs. (12))
d = root-x coefficient for T (Eqs. (12))
$\mathbf{C}=1/R$ = curvature (Eq. 5 and Figs. 4*b* and 5*b*)
C_L = tangential-loading coefficient (Eqs. (7) and (8))
e = coefficient of the construction lines (Eq. 11)
g = body force (Eqs. (2) and (3))
M = Mach number
p = pressure (Eqs. (2) and (3))
P_f, P_m, $P2$ = surface-segment nodes
R = local radius of curvature (Eq. (5))
s = blade-surface length (Fig. 10)
S = tangential spacing of the blades (Fig. 4*a*)
t = time (Eqs. (1), (2) and (3))
T = thickness distribution (Eq. (12))
v_i = velocity component ($i=r, \theta$)
$\mathbf{v}$ = velocity vector (Eq. 4)
(x, y) = Cartesian coordinates
(X, Y) = nondimensional distances ($X=x/b$, $Y=y/b$)
$y1$, $y2$ = blade-surface approximations (Eq. (10))
α = flow angle
β = angle of the blade surface
δ_1 = boundary-layer displacement thickness (Figs. 5*c* and 10)
λ = stagger angle of the blades (Fig. 4*a*)
μ = dynamic viscosity (Eq. 4)
ρ = density (Eqs. (1), (2), and (3))
τ_{ij} = shear stress components (Eq. (4), $i, j=r, \theta$)
ϕ = angle of throat diameter with x axis (Fig. 4*a*)
ω = thickness distribution parameter (Eqs. (11) and (12))

Subscripts

1, 2 = point P, flow angle or Mach number, inlet and outlet, respectively
f = segment knot near the leading edge (Figs. 4*a* and 4*c*)
m = segment knot near the trailing edge (Fig. 4*a*)
p = pressure side
r = radius in polar coordinates (r, θ)
s = suction side
θ = angle in polar coordinates (r, θ)
κ = construction line

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distributions along the blade surfaces require smooth surface-curvature distributions (continuous slopes of velocity and curvature along the blade surface). Continuous slope of curvature requires continuous derivatives, as illustrated in the following two equations for curvature and slope of curvature for $y=f(x)$, $y'=df(x)/dx$, $y''=d^2f(x)/dx^2$ and $y'''=d^3f(x)/dx^3$:

$$\mathbf{C}=\frac{1}{R}=\frac{y''}{[1+y'^2]^{(3/2)}} \tag{5}$$

$$\mathbf{C}'=\frac{d\mathbf{C}}{dx}=\frac{y'''[1+y'^2]-3y'y''^2}{[1+y'^2]^{(5/2)}} \tag{6}$$

Experimental Evidence. Most parametric splines currently in use for direct blade design (e.g., cubic, B-splines, Bezier splines, etc.) have continuous first and second derivatives and they result in smooth-looking surfaces with continuous curvatures, but discontinuous slopes of curvature (y''' in Eq. (6)) at the spline knots. One must distinguish here between the small local slope-of-curvature discontinuities from surface roughness and fouling, with which turbines must operate, and the slope-of-curvature discontinuities in the as-designed shape at the junctions of the splines. It will be shown below that the latter are invisible to the eye, but they may produce unusually loaded cascades and thicker wakes. This has resulted in many test and production airfoils (isolated and in turbomachine cascades) that exhibit spikes or dips of various magnitudes in Mach number and pressure-coefficient distribution, which occasionally result in unexpected loading distributions along the blade length, and in local separation bubbles. These effects are visible as small local "kinks" in surface pressure or Mach number distributions in some of the computational and experimental data published, for example, by: Okapuu (1974), Gostelow (1976), Wagner et al. (1985), Hourmouziadis et al. (1987) (with separation bubble), Sharma et al. (1992) and others.

Blending a leading-edge circle or ellipse with the blade surfaces usually results in a local curvature or slope-of-curvature discontinuity, which may cause separation and substantial leading-edge surface Mach number spikes (detected only if the probes are located at the correct location). Such a local leading-edge laminar-separation bubble due to blending of a leading-edge circle with the blade surfaces occurs in the turbine geometry published by Hodson and Dominy (1987), seen in the data published in Fig. 11 of Hodson (1985). This leading-edge spike and separation region were removed by modifying the geometry of the blade in the vicinity of the slope-of-curvature discontinuity with an inverse design technique as explained in Figs. 11, 12, and 13 of Stow (1989).

The full extent of the effects of slope-of-curvature discontinuities on blade performance, losses, and boundary-layer behavior is still under investigation. While it may be possible intentionally to trip the flow into turbulence by imposing a local slope-of-curvature discontinuity, it is clear that a prescribed-curvature-distribution blade-design method is a useful tool.

Accuracy of Computations

The performance of sample turbine blades in this paper has been evaluated using the inviscid and steady-flow version of Giles' (1988) computer program UNSFLO. The accuracy of the computations was checked by comparing the results of Euler computations with three cases of known output. The first case was a comparison of computations with the (exact) theoretical solution of Gostelow's (1984) compressor cascade. For this case $M_1=0.200$, $M_2=0.135$, $\alpha_1=53.5$ deg and $\alpha_2=30.0$ deg. A fine uniform H grid of 400×100 was used to resolve the flow around the leading edge. The results, shown in Fig. 1, indicate excellent agreement. The tick marks are the analytic solution and the straight line is the calculation by UNSFLO.

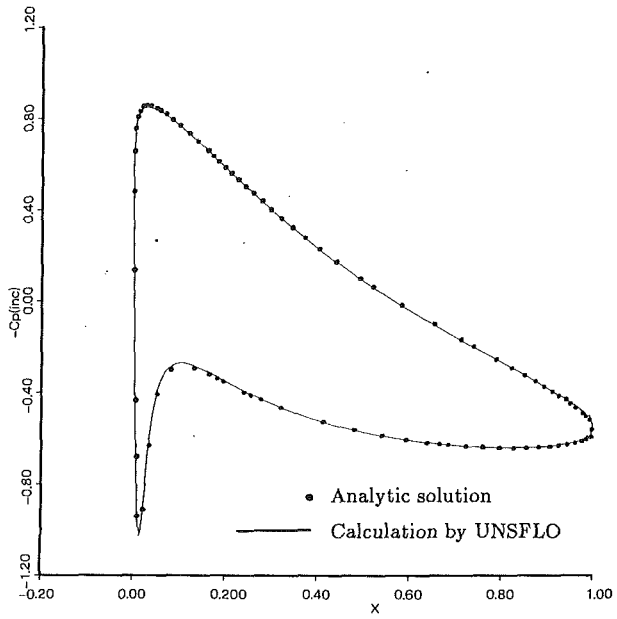


Fig. 1 Comparison of incompressible lift coefficient for Gostelow's (1984) compressor cascade

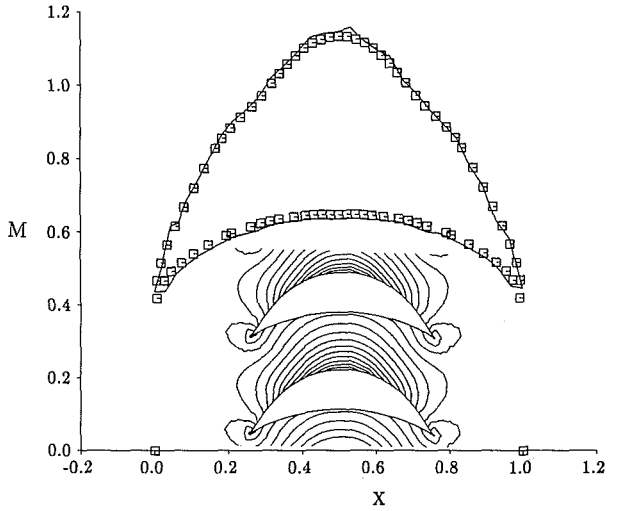


Fig. 2 Comparison of surface mach numbers for Hobson's turbine cascade No. 2 (from AGARD, 1985)

The second case was a comparison of computations on uniform H grid with the hodograph solution of Hobson's transonic turbine cascade No. 2 (taken from AGARD, 1985). This is a singular case; small perturbations result in shocks in the passage. The inlet and exit flow angle and Mach number are 46.123 deg and 0.575, respectively. A relatively coarse grid of 60×15 was used. The results, shown in Fig. 2, also indicate excellent agreement. The tick marks are the hodograph solution and the straight line is the calculation by UNSFLO.

The first two cases were suitable for inviscid-flow checks with sharp trailing edges, and indicate the accuracy of the computations. The following experimental cases, coupled with the effect of surface curvature on the performance of the sample cascades shown below, indicate the importance of surface-curvature distribution, and the need for it to be taken into account in blade-design methods.

The third case is a comparison of inviscid computations with the experimental results from the subsonic turbine cascade obtained at DFVLR Braunschweig by Kiock et al. (1986). For this case $M_1=0.260$, $M_2=0.782$, $\alpha_1=30.0$ deg and $\alpha_2=-67.33$

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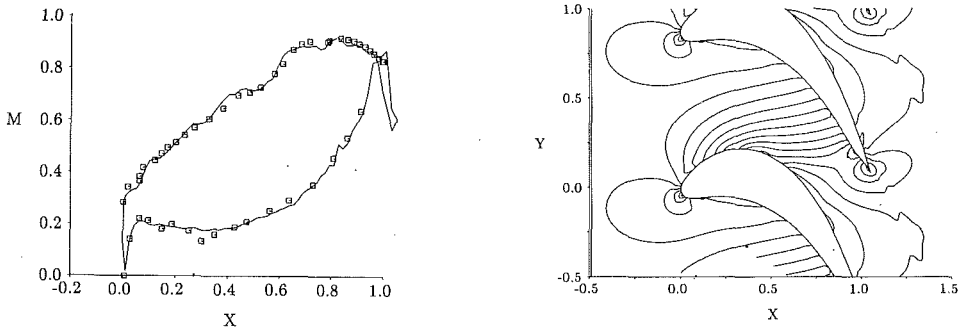


Fig. 3 Comparison of Mach number distributions for the turbine cascade investigated by Kiock et al. (1986): left, surface Mach-number distributions: Right, Mach number contours of increment 0.05

deg. A uniform H grid of 100×40 and a cusp (to close the trailing edge) were used. The results, shown in Fig. 3, indicate very good agreement with the experiments. The tick marks are the experimental results (transformed from data along percentage of chord c to data along percentage of axial chord b, along X) and the straight line is the calculation. Despite the inviscid nature of the computation, the numerical solution follows local dips in the suction-surface Mach number distribution at $X\approx0.55$ and $X\approx0.80$, and local accelerations at $X\approx0.05$ and $X\approx0.70$. Korakianitis and Papagiannidis (1993a) show an additional check with very good agreement between computation and experiment for the turbine geometry tested by Hodson and Dominy (1987). In the latter case their experiment and our computation indicate a leading-edge Mach number spike on the suction surface. The spike is caused by a curvature or slope-of-curvature discontinuity in blending the leading-edge circle with the blade surface. The comparisons with experiments and the effect of curvature discontinuities discussed below indicate the importance of curvature distribution along the blade surface on performance.

Cascade-Design Parameters

Simplified examples of geometric blade-design sequences are described in some texts such as Wilson's (1984). The designer has many choices, but for the purposes of this paper we will refer to the axial-turbine-cascade diagram shown in Fig. 4(*a*). We will assume that the designer has chosen the blade row velocity diagram and its parameters: the inlet flow angle α_1; the outlet flow angle α_2; and the exit Mach number M_2 (in the absolute frame for stators and in the relative frame for rotors). Four additional choices must be made: the nondimensional tangential spacing S/b between the blades; the stagger angle λ; the nondimensional throat diameter o/b; and the trailing-edge thickness.

Some guidelines for the stagger angle λ for low-speed turbine blades have been published by Kacker and Okapuu (1982), although the designer has considerable flexibility in choosing λ. It will be shown below that the value of λ also dictates whether the cascade is front or aft loaded.

The spacing between the blades (which is directly linked to the number of blades in the blade row) is a function of the tangential lift coefficient C_L, defined by Zweifel (1945):

$$C_L \equiv \frac{\textit{tangential aerodynamic force}}{\textit{tangential blade area}\times\textit{outlet dynamic head}} \tag{7}$$

This expression can be manipulated in a number of ways (for compressible flow, for incompressible flow, accounting for variations in axial-flow velocity, etc.). Here we chose the incompressible-flow derivation (see Wilson, 1984), which reduces C_L to:

$$C_L = 2\frac{S}{b}\cos^2\alpha_2(\tan\alpha_1-\tan\alpha_2) \tag{8}$$

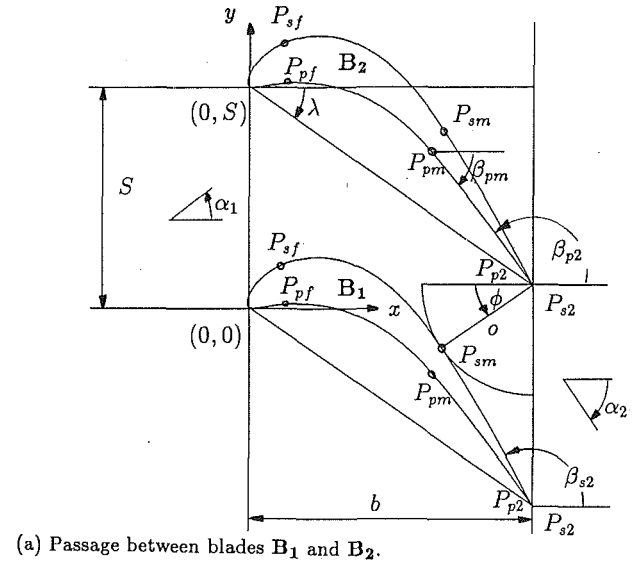
The value of C_L is determined by the number of blades in the cascade, which dictates S. Typically $0.8<C_L<1.2$; earlier designs had lower values of C_L, and in modern cascades C_L has been gradually increasing. It is difficult to design high-efficiency cascades that have both high values of C_L and high values of deflection ϵ in the cascade, where $\epsilon=\alpha_1-\alpha_2$.

The throat diameter is an extremely important design input because it dictates the mass flow that can be passed through the cascade, and hence the work that can be delivered by the turbine. A good first approximation is $o/S=\cos^{-1}\alpha_2$. Experimental figures linking the throat diameter o to the blade spacing S, the throat Mach number M_2, and the curvature of the convex (suction) blade surface near the trailing edge have been published by Ainley and Mathieson (1951), and are used extensively in the early stages of blade design. The industry has for years relied on empirical relationships based on the Ainley-Mathieson data and on improvements to it (e.g., by Dunham and Came, 1970) for the prediction of α_2 as function of M_2, S and o. These empirical data have been simplified by others for design applications (for example by Wilson, 1984, and Dunham, 1974). This is sufficient for preliminary design. Today's computer programs are more accurate and compute the outlet flow angle for a given geometry. Gostelow (1975) has explained how the result depends not only on the geometry, but also to some extent on the exact location of the point at which the Kutta condition is applied at the trailing edge.

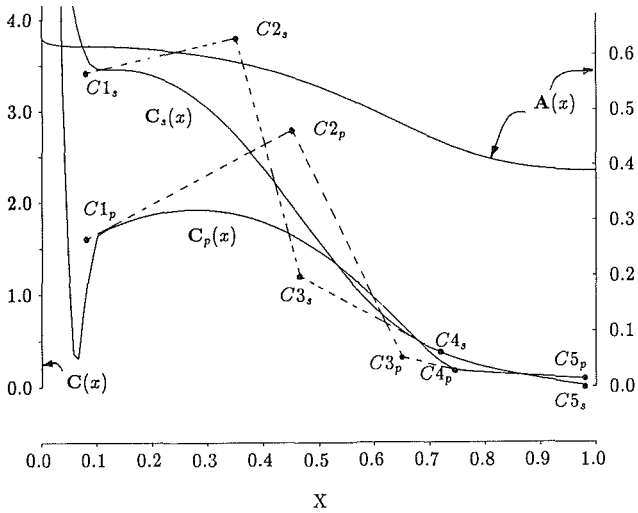
The trailing-edge thickness should be as small as manufacturing and strength considerations would allow (to minimize the wake incident on the next blade row). It is also affected by geometric constraints imposed by the cooling slots in cooled blades. In the sample airfoils of this paper the trailing edge is a point, because we have used an inviscid flow solver to analyze the blade performance. But the method presented enables the designer to specify a trailing-edge thickness, which can be filled with a sharp wedge, cusp, trailing-edge circle, or other appropriate shape.

One would like the velocity of the fluid adjacent to the concave (pressure) surface to accelerate from the stagnation point near the leading edge to the trailing edge. An exception to this are the low-pressure stages of turbines, where in some cases a local decrease in velocity between the leading edge and midchord is designed to increase loading. Continuous acceleration is not as easy to accomplish for the velocity of the fluid adjacent to the convex (suction) surface, because the flow is "unguided" on the suction surface downstream of the throat in what is called the region of "unguided diffusion." The velocity of the fluid adjacent to the convex (suction) surface accelerates from the stagnation point near the leading edge to some point near the throat, and then it usually decelerates. Less deceleration (diffusion) means better designs. If the flow decelerates too much, then the boundary layer will separate, the blade wake will be thick, and the resulting cascade will have poor performance (low efficiency).

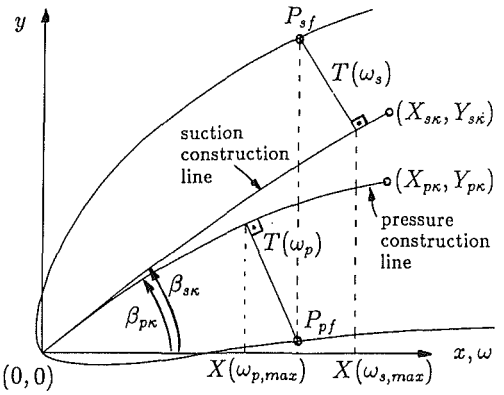
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(a) Passage between blades $\mathbf{B_1}$ and $\mathbf{B_2}$.



(b) Curvature $\mathbf{C}(x)$ and area $\mathbf{A}(x)$ distributions

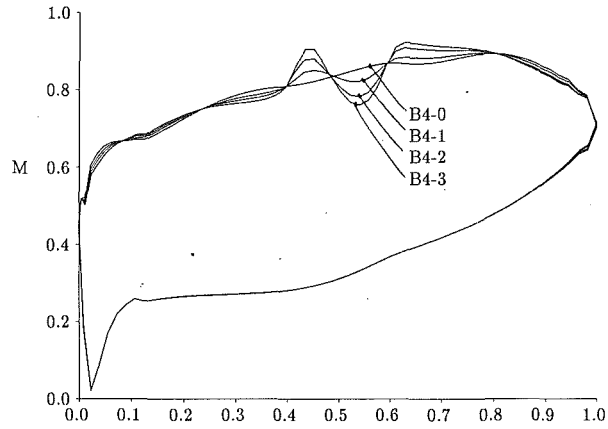


(c) Leading edge thickness distributions

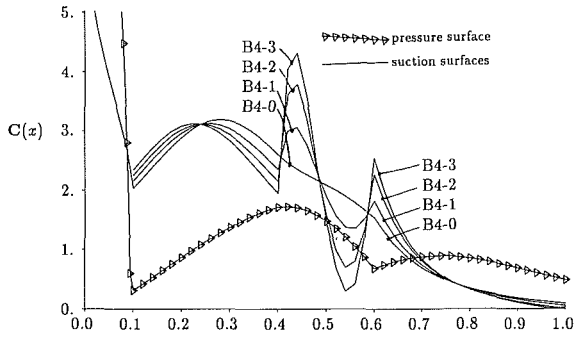
Fig. 4 Geometry of a typical cascade, and parameters of the blade-design method

The Blade-Design Method

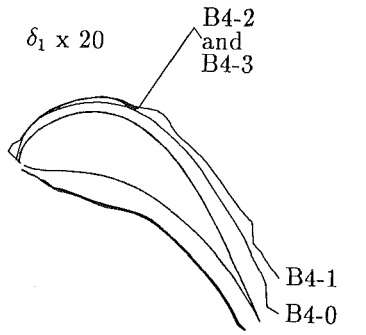
In past investigations we introduced a blade design method in which the blade surfaces were prescribed by analytic polynomials $y=f(x)$, (Korakianitis, 1989). In that method the throat diameter and its location were an output of the location of the two surfaces. The resulting blade performance exhibited a dependence on curvature, which is illustrated in Fig. 5 for a turbine cascade designed for $\alpha_1=40$ deg, $\alpha_2=-60$ deg,



(a) Mach number distributions



(b) Curvature $\mathbf{C}(x)$ distributions



(c) Effect on boundary layers (using Albers and Gregg's (1974) program with freestream turbulence 10%)

Fig. 5 Surface distributions for cascades B4-0, B4-1, B4-2, B4-3: $\alpha_1=40$ deg, $\alpha_2=-60$ deg, $C_L=1.00$, $\lambda=-32$ deg

$M_2=0.8$, and $C_L=1.0$. Figure 5(*a*) shows the Mach number distributions of four blades of almost identical geometries, and Fig. 5(*b*) shows the corresponding surface curvature distributions. Figure 5(*c*) shows the boundary-layer displacement thickness $\delta_1\times 20$ of each blade calculated with Albers and Gregg's (1974) differential boundary layer program assuming incoming free-stream turbulence of 10 percent. All four blades have been plotted on each other in Fig. 5(*c*). The pressure surfaces are identical. The suction surfaces have differences that are small enough not to be visible in the figure. Blade B4-0 has smooth curvature and Mach number. Blade B4-1 has a small discontinuity in the slope of curvature that causes a disturbance in surface Mach number distribution, and a thicker boundary layer (and wake) at the trailing edge. Blades B4-2 and B4-3 have correspondingly larger slope-of-curvature discontinuities that cause the boundary layer to separate near the discontinuity.

A second blade design method with parametric splines in ξ ($x=f(\xi)$ and $y=f(\xi)$) resulted in a very cumbersome and time-consuming design sequence (Korakianitis and Pantazopoulos,

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1993b). Through these efforts it became apparent that we wanted to control the curvature distribution as well as the location of the blade surfaces. This was accomplished by the design method presented below.

The Trailing-Edge Region. Typically the Mach number at the outlet-flow plane, M_2, will be a design specification. The values of α_1, α_2, and C_L also specify S/b; the throat diameter $o/S = \cos\alpha_2$ as described above. Choosing the value of λ and the trailing edge thickness specifies the trailing edge points P_{s2} and P_{p2} (these are shown at the same point in Fig. 4(*a*) since the designs will be analyzed with an inviscid flow program).

The trailing edge region of the suction surface is specified by a third-order polynomial $y = f(x)$ evaluated using four boundary conditions: the location of P_{s2}; the blade angle β_{s2} at the trailing edge; and the angle ϕ of the throat-circle diameter with the x axis. The latter specifies two boundary conditions: the location of point P_{sm}, and the tangent of the third-order polynomial at that point, equal to $(\pi/2 + \phi)$. The analytic polynomial has continuous first, second, and third derivatives. With these we compute the curvature of the resulting blade surface $\mathbf{C}_s(x)$ (using Eq. (5)) between points P_{sm} and P_{s2}, and plot it between points $C4_s$ and $C5_s$ in Fig. 4(*b*).

The trailing edge region of the pressure surface is specified by a third-order polynomial $y = f(x)$ evaluated using four boundary conditions: the location of P_{p2}; the blade angle β_{p2} at the trailing edge; the (x, y) coordinates of a point P_{pm} on the surface; and the angle β_{pm} of the blade surface at point P_{pm} with the x axis. With the first and second derivatives of this polynomial we compute the curvature of the resulting blade surface $\mathbf{C}_p(x)$ (using Eq. (5)) between points P_{pm} and P_{p2}, and plot it between points $C4_p$ and $C5_p$ in Fig. 4(*b*).

The slopes of the curvatures $\mathbf{C}_s(x)$ and $\mathbf{C}_p(x)$ at points $C4_s$ and $C4_p$, respectively, are computed and become inputs for the rest of the design. A gradually increasing curvature distribution in the trailing edge regions results in mechanically and aerodynamically feasible blade shapes. Different (smooth) curvature distributions would be required for supersonic blades.

The Main Part of the Blades. The design is converted into an effort to specify a curvature distribution for the shape of the blade surfaces $\mathbf{C}_s(x)$ and $\mathbf{C}_p(x)$. In order to ensure slope-of-curvature continuity, the curvature from points P_f to points P_m is specified using a Bezier spline in curvatures using the following conditions (see Fig. 4(*b*). On the curvature of the suction surface we specify points $C1_s$, $C2_s$, and $C3_s$. Point $C4_s$ is already specified from the corresponding value of curvature, evaluated from the trailing-edge third-order polynomial. The x location of point $C3_s$ is variable, but the value of curvature there is evaluated such that line $C3_sC4_s$ is tangent to the surface-curvature line at point $C4_s$. Point $C2_s$ is user specified. Point $C1_s$ is specified at an x location corresponding to P_f, or at an x location just ahead of P_f. Since the slope of the Bezier curve is tangent to the line of knots at its ends, the tangency condition at point $C4_s$ ensures slope-of-curvature continuity from $C1_s$ to $C5_s$ (from P_{sf} to P_{s2}). Similarly points $C1_p$, $C2_p$, and $C3_p$ are specified on $\mathbf{C}_p(x)$.

Using central differences Eq. (5) is written as:

$$\mathbf{C}_i = \frac{CF1/CF2}{CF3}$$

$$CF1 = 2\left(\frac{y_{i+1}-y_i}{x_{i+1}-x_i} - \frac{y_i-y_{i-1}}{x_i-x_{i-1}}\right) \tag{9}$$

$$CF2 = (x_{i+1}-x_{i-1})$$

$$CF3 = \left\{\left[\frac{1}{2}\left(\frac{y_{i+1}-y_i}{x_{i+1}-x_i} + \frac{y_i-y_{i-1}}{x_i-x_{i-1}}\right)\right]^2 + 1\right\}^{3/2}$$

From the above equation, given (x_{i-1}, y_{i-1}), (x_i, y_i), x_{i+1}, and $\mathbf{C}_i$ we can compute y_{i+1}. This can be done by manipulating Eq. (9) into a sixth-order linear algebraic equation in y_{i+1}, or by a numerical solution. We have implemented a regula-falsi solution for y_{i+1} using Eq. (9). This is solved starting from points P_m and progressing explicitly to points P_f. The Bezier spline is iteratively manipulated until the slope and the y location of the blade surfaces at points P_f, and the shapes of the curvature distributions are acceptable.

In some cases, trying to make the blade surfaces pass with desirable slopes through desirable P_f points becomes tedious. For those cases we provide an alternative solution. The designer specifies the (x, y) location of point P_f. A parabola passing through points P_f, P_m, and the next point on the blade surface in the trailing-edge region in used as a first approximation $y1(x)$ to the blade surface. Equation (9) is written as:

$$[y2_{i+1} - 2y2_i + y2_{i-1}] = \Delta x^2 \mathbf{C}_i \left[\left(\frac{y1_{i+1}-y1_{i-1}}{2\Delta x}\right)^2 + 1\right]^{3/2} \tag{10}$$

Equation (10) is written as a tridiagonal matrix in $y2_i$. Points P_m and the input points for P_f are incorporated as boundary conditions in the first and last rows of the right-hand side. The system of equations is solved with an LU decomposition. The values of $y1_i$ on the right-hand side are set equal to the previous solution for $y2_i$, and the process is repeated until the values of $y1_i$ and $y2_i$ converge to the same solution (after about twelve iterations). The converged blade surface in the final $y2_i$ passes through points P_f and P_m, but in the general case results in a spike in curvature at point P_m. The curvature input points $C1$, $C2$, and $C3$ are iteratively manipulated until the spike is minimized, and then having found the inputs that will make the blade surface pass through point P_f, the designer returns to specifying the blade surface by solving Eq. (9).

Leading-Edge Geometry. The leading edge geometry is designed by specifying thickness distributions added perpendicularly to parabolic construction lines that pass through the leading edge, as shown in Fig. 4(*c*). The parabolic construction lines are specified by the angle of the construction line at the origin β_κ, the (x, y) location of the origin (0, 0), and of one additional point (X_κ, Y_κ) on the construction line. The parabolic construction lines define a parameter ω along x, and the corresponding base line $P(\omega)$ as follows:

$$\begin{aligned} x &= \omega \\ P(\omega) &= e_0 + e_1\omega + e_2\omega^2 \end{aligned} \tag{11}$$

where the constants e_0, e_1, and e_2 are evaluated from the boundary conditions of the parabolic construction lines. For the suction line we add a thickness distribution above the parabolic construction line, and for the pressure line we add a thickness distribution below the parabolic construction line. The thickness distribution is of the form:

$$\begin{aligned} T_s(\omega) &= d_s\sqrt{\omega} + c_{s1}\omega + c_{s2}\omega^2 + c_{s3}\omega^3 + c_{s4}\omega^4 \\ T_p(\omega) &= d_p\sqrt{\omega} + c_{p1}\omega + c_{p2}\omega^2 + c_{p3}\omega^3 + c_{p4}\omega^4 \end{aligned} \tag{12}$$

where d is specified by the designer. The coefficients c_1, c_2, c_3, and c_4 are derived with the conditions that the absolute values, first, second, and third derivatives of the thickness distribution and of the downstream blade segment at point P_f with respect to the parabolic construction line are equal. This ensures curvature and slope-of-curvature continuity at point P_f, and avoids the need to blend a circle, ellipse, or other shape with the blade surfaces near the leading edge. The continuous slope of curvature at point P_f also ensures that the blade will not exhibit unexpected leading-edge separation bubbles. The value of parameter d can be used to specify thicker or thinner leading edges, which would affect the design-point and off-design-point operating performance of the cascade. These are the subject of a future investigation. The origin (0, 0) itself is

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Table 1 Parameters specifying blades L25, L30, L35, and L40

Blade Parameter	L25	L30	L35	L40
α_1	40.00	40.00	40.00	40.00
α_2	-60.00	-60.00	-60.00	-60.00
C_L	1.00	1.00	1.00	1.00
o/S	0.5000	0.5000	0.5000	0.5000
λ	-25.00	-30.00	-35.00	-40.00
$\beta_{s\kappa}$	40.0000	40.0000	40.0000	40.0000
$XS_{s\kappa}$	0.1000	0.2000	0.2000	0.3300
$YS_{s\kappa}$	0.1700	0.0500	0.1300	0.0500
d_s	0.0800	0.1000	0.1500	0.1500
β_{s2}	-64.5000	-65.0000	-62.0000	-62.500
ϕ	38.5000	38.0000	34.5000	33.5000
$x(C1_s)$	0.0800	0.0800	0.08000	0.08000
$y(C1_s)$	2.9000	3.0000	3.4200	4.6000
$x(C2_s)$	0.3400	0.4000	0.3500	0.2300
$y(C2_s)$	3.7500	3.0000	3.8000	3.3000
$x(C3_s)$	0.6865	0.6000	0.4650	0.5200
$\beta_{p\kappa}$	40.0000	40.0000	40.0000	40.000
$XS_{p\kappa}$	0.2000	0.2000	0.2000	0.3000
$YS_{p\kappa}$	0.0900	0.4100	0.1000	0.0200
d_p	0.0500	0.1000	0.1500	0.2000
β_{p2}	-50.5000	-55.0000	-55.0000	-56.7000
β_{pm}	-38.7000	-47.0000	-52.0000	-52.3000
$x(P_{pm})$	0.7500	0.7500	0.7500	0.7400
$y(P_{pm})$	-0.2180	-0.2650	-0.3600	-0.4700
$x(C1_p)$	0.0800	0.0800	0.0800	0.0800
$y(C1_p)$	0.8000	0.8000	1.6000	2.0000
$x(C2_p)$	0.5600	0.5100	0.4500	0.3850
$y(C2_p)$	2.2000	2.9000	2.7900	2.3000
$x(C3_p)$	0.5500	0.6500	0.6500	0.4300

a singular point. In order to achieve first-derivative continuity at that point, the angles of the parabolic construction lines should be equal for the suction and pressure surfaces.

Some Features of the Method. This blade-design method has been used to design numerous cascades of various loading distributions. Through these design sequences it was found that the designer has a lot of freedom in specifying the curvature (and location of the blade surface) in cascades of medium and low loading (C_L) (Korakianitis and Papagiannidis, 1993a). Blades with high turning and high values of C_L require very fine adjustments in blade curvature (and location) in order to attain good performance. This is achieved by modifying the above method to use more Bezier splines to specify the curvature in Fig. 4(*b*).

In using the method we have found it useful to keep the location of points P_f at $X=0.08$. After very few geometric-design iterations, the user finds the range of values of curvatures at points $C1$ that give acceptable geometries. These values are between 2 and 6 on the suction surface, and about 1 on the pressure surface. The same leading-edge parabolas can be specified by the same angle β_κ and any of an infinite number of points (X_κ, Y_κ). We have found it useful to keep $X_\kappa=0.20$ to facilitate understanding of the effect of changing the location of Y_κ on leading-edge geometry (local curvature distribution). The design method is easy enough to master that a user improved the initial design of sample cascades L25 and L40 (see below) before he realized this intention (see Table 1).

We have also found it useful to keep the x location of point P_{pm} near the x location of point P_{sm} to control the surface angles and the wedge angle between the surfaces (inputs $\phi+\pi/2$ and β_{pm}) at that region of the airfoil. Similarly β_{s2} and β_{p2} are used to specify the trailing-edge wedge angle (it becomes an input). The curvature distribution results in (x, y) points near P_{sf} and P_{pf} that give as output the wedge angle near the leading edge. The surface-curvature distributions are manipulated until desirable wedge angles are obtained. The surface-angle distributions are part of the graphic output during use of the method.

The exact value of curvature and location of points C_1, C_2, C_3, and C_4 is not as critical as the resulting shape of the curvature distribution. In using the method we compare desired changes in Mach number distribution with changes in *curvature distribution* and location of blade surface. The alternative method (Eq. (10)) allows one to match up the leading edge position, thickness, and wedge angle in any region with the rest of the design (near the trailing edge).

After the first iteration (first geometric design and analysis), the user examines the resulting loading distribution, and appreciates where to increase and decrease local curvature (and local loading). After the second iteration the user gains an appreciation of the magnitude of the required changes in curvature to cause the desired changes in Mach number distribution. The procedure is repeated until a desirable geometry and performance is obtained (as in the case of the sample cascades below), or until the designer decides that the shape is good enough to proceed to more advanced (inverse) design methods.

Effects of Stagger Angle

The performance of four cascades designed with the prescribed-curvature distribution method is shown in Figs. 6, 7, 8, and 9. For all cases the main cascade-design variables except the stagger angle are kept constant (for example $\alpha_1=40$ deg, $\alpha_2=-60$ deg, $M_2=0.800$, $C_L=1.00$, and $o/S=0.5$). The stagger angles of cascades L25, L30, L35, and L40 are $\lambda=-25$ deg, $\lambda=-30$ deg, $\lambda=-35$ deg, and $\lambda=-40$ deg respectively. Other design inputs are shown in Table 1. In all cases the computational grid was 100×40. These figures illustrate that increasing the stagger angle, while keeping other design parameters constant, results in thinner and more front-loaded cascades. This is because as λ is increased the throat circle pushes both blade surfaces farther down, thus locating the throat farther upstream on the suction surface and thinning the blade. In order to fit the rest of the blade into the passage the designer is forced to increase the curvature of the suction surface of the blade in the vicinity of point P_{sf} (as shown in Table 1 $y(C1_s)$ is increasing with stagger angle). The increased curvature in turn increases the Mach number in that vicinity, thus loading the front part of the suction surface of the blade accordingly. While it is relatively easy to change the type of loading distribution of cascade L25 (Korakianitis and Papagiannidis, 1993a), it is harder to change the front-loaded cascade L40.

The relatively aft-loaded cascade of Fig. 6 ($\lambda=-25$ deg) has a relatively short length of relatively very adverse pressure gradient. As the stagger angle is increased, this changes until the relatively front-loaded cascade of Fig. 9 ($\lambda=-40$ deg) has a relatively long length of relatively not as adverse pressure gradient. This clearly affects the development of the pressure and suction-surface boundary layers. In a previous investigation Hoheisel et al. (1987) found that a midloaded cascade had a relatively thinner boundary layer, although their blades were designed with inverse-design methods and did not have the rest of the design parameters identical. A similar trend is shown in Fig. 10, which shows the pressure-side and suction-side boundary-layer displacement thickness $\delta_1\times20$ along the nondimensionalized blade-surface length s/b, calculated for the four cascades L25, L30, L35, and L40, using Albers and Gregg's (1974) program, for 10 percent free-stream turbulence. Blade L35 (with intermediate loading distribution) has the thin-

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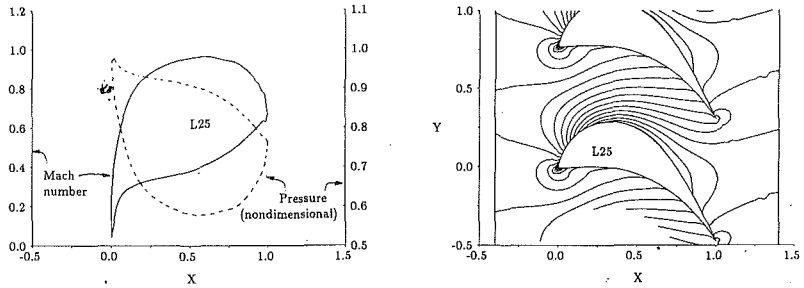


Fig. 6 Performance of cascade L25 (stagger angle λ = 25 deg); blade-design parameters shown in Table 1: left, surface Mach number and nondimensional surface pressure distributions; right, Mach number contours (increment 0.05)

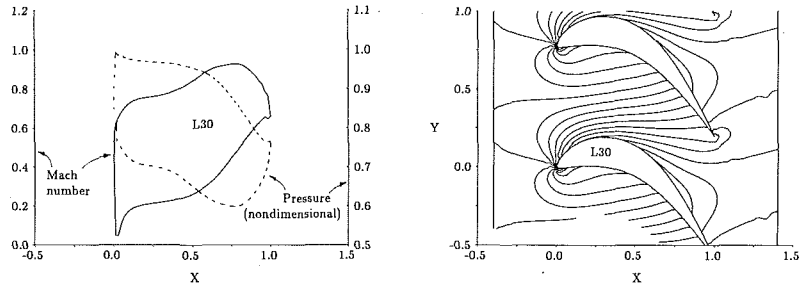


Fig. 7 Performance of cascade L30 (stagger angle λ = − 30 deg); blade-design parameters shown in Table 1: left, surface Mach number and nondimensional surface pressure distributions; right, Mach number contours (increment 0.05)

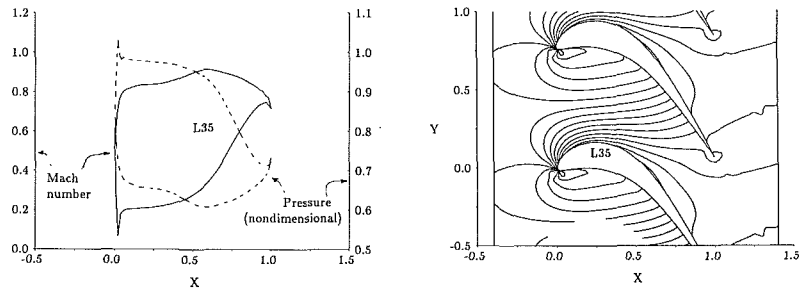


Fig. 8 Performance of cascade L35 (stagger angle λ = − 35 deg); blade-design parameters shown in Table 1: left, surface Mach number and nondimensional surface pressure distributions; right, Mach number contours (increment 0.05)

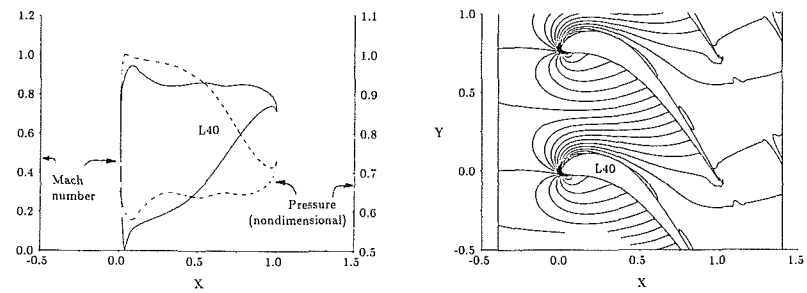


Fig. 9 Performance of cascade L40 (stagger angle λ = − 40 deg); blade-design parameters shown in Table 1: left, surface Mach number and nondimensional surface pressure distributions; right, Mach number contours (increment 0.05)

nest boundary layer thicknesses on both surfaces and the thinnest wake.

Conclusions

Small slope-of-curvature discontinuities in the blade surfaces result in Mach number or pressure-coefficient surface-distribution spikes or humps, which may introduce regions of local acceleration and deceleration, separation, or undesirable loading distributions along the length of the blades. A two-dimensional direct blade-design method has been developed. It uses a mixture of analytic polynomials and a mapping of a desired curvature distribution on the (x, y) plane to specify the geometry of the cascades. By specification the blade surfaces have continuous slope of curvature throughout their length. The method permits the user to specify the leading edge by two thickness distributions around two independent construction lines, in a manner that avoids the overspeed regions near the leading edge (because it does not employ the usual blending in of the curvatures near the "leading-edge circle or ellipse"). It can be used to design subsonic or supersonic blades for

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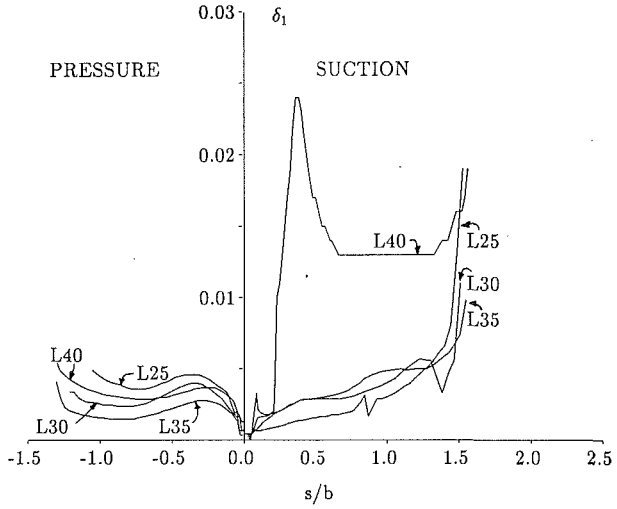


Fig. 10 Boundary layer displacement thickness δ_1 for cascades L25, L30, L35, and L40 (using Albers and Gregg's (1974) program with free-stream turbulence 10 percent)

compressors or turbines, or isolated airfoils, but the discussion in this paper is limited to high-Mach-number subsonic-exit turbine blades.

It is found that the cascade performance, judged by the shape of the surface Mach number distribution, is extremely sensitive to small changes of the surface geometry and of the curvature distribution of the airfoils. The design method presented ensures that the boundary layer, which is usually transitional and turbulent along the blade surfaces, is not perturbed unintentionally. The smooth curvature and Mach number distributions also enable the user to design lower-loss higher-loaded cascades. Continuous-slope-of-curvature surfaces should be used wherever the aerodynamic, heat transfer, hydrodynamic, or aesthetic performance, appropriately defined in each case, is critical.

The performances of four cascades of different stagger angles, but otherwise identical blade-design inputs and design criteria, have been computed, and their boundary-layer behavior has been numerically investigated. It is concluded that increasing the stagger angle results in more front-loaded cascades. An intermediate-loading philosophy (at an intermediate stagger angle) results in thinner wakes.

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